

IMPLEMENTATION OF PAGE REPLACEMENT ALGORITHM

Ex. No. :4

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AIM

To implement FIFO page replacement algorithm and count page faults.

PROCEDURE

1. Enter the page reference string.
2. Enter the number of frames.
3. Load pages into frames using FIFO method.
4. Count and display page faults.

PROGRAM

```
pages = list(map(int, input("Enter page reference string: ").split())) frames_count = int(input("Enter number of frames: "))
frames = [] index = 0 faults = 0
```

```
for page in pages:
```

```
    if page not in frames:
```

```
        if len(frames) < frames_count:
```

```
            frames.append(page)
```

```
        else:
```

```
            frames[index] = page
```

```
index = (index + 1) %
```

```
frames_count faults += 1
```

```
print("Frames:", frames)
```

```
print("Total Page Faults =", faults)
```

OUTPUT

Practice 4 - Sample Output

```
python output
Enter page reference string: 1 2 3 2 4 1 5
Enter number of frames: 3

Frames: [1]
Frames: [1, 2]
Frames: [1, 2, 3]
Frames: [1, 2, 3]
Frames: [4, 2, 3]
Frames: [4, 1, 3]
Frames: [4, 1, 5]

Total Page Faults = 6
```

RESULT

Thus, the given programs were implemented successfully.

The expected outputs were obtained and the required concepts were verified.